

Design and FPGA Implementation of a Digital Frequency-to-Voltage Converter Using an I/Q Frequency Discriminator

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Abstract

This report describes the design and FPGA implementation of a digital frequency-to-voltage (F-to-V) converter using a Red Pitaya STEMLab 125-14. The objective is to recover rapid frequency variations around a nominal 20 MHz carrier and represent those variations as an analog voltage.

An initial implementation based on a 16-period reciprocal frequency counter was suitable for measuring approximately stationary frequencies, but its finite measurement window caused averaging and latency when the input frequency changed rapidly. To overcome this limitation, a digital quadrature frequency discriminator was implemented.

The new architecture performs digital I/Q downconversion using a 20 MHz numerically generated reference, low-pass filtering, complex-conjugate phase discrimination, phase-to-frequency conversion, and frequency-to-voltage scaling. The resulting voltage is sent through the Red Pitaya DAC. The implemented system is designed such that a frequency deviation of approximately ± 100 kHz around 20 MHz corresponds to approximately ± 1 V.

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1 Objective

The objective of the system is to convert instantaneous frequency deviation around a carrier frequency into an analog voltage.

The input frequency can be written as

$$f_{\text{in}}(t) = f_c + \Delta f(t), \quad (1)$$

where

$$f_c = 20 \text{ MHz} \quad (2)$$

is the nominal carrier frequency and $\Delta f(t)$ is the time-dependent frequency deviation. The desired frequency-to-voltage relationship is

$$V_{\text{out}}(t) = K_{FV} \Delta f(t), \quad (3)$$

where the chosen sensitivity is

$$K_{FV} = 10 \text{ pV/Hz}. \quad (4)$$

Therefore,

$$19.9 \text{ MHz} \rightarrow -1 \text{ V}, \quad (5)$$

$$20.0 \text{ MHz} \rightarrow 0 \text{ V}, \quad (6)$$

$$20.1 \text{ MHz} \rightarrow +1 \text{ V}. \quad (7)$$

Consequently, frequency modulation around the 20 MHz carrier can be observed directly as a voltage waveform on an oscilloscope.

2 Hardware Platform

The implementation was developed for the Red Pitaya STEMLab 125-14.

The relevant hardware characteristics are:

Table 1: Relevant Red Pitaya parameters.

Parameter	Value
ADC resolution	14 bit
ADC sampling frequency	125 MS/s
DAC resolution	14 bit
DAC sampling frequency	125 MS/s
FPGA device	Xilinx Zynq-7010
FPGA clock used	125 MHz
Input used	IN2
Output used	OUT2
Nominal carrier	20 MHz

The complete conversion is performed digitally inside the FPGA after the input signal is sampled by the ADC.

3 Initial Reciprocal Counter Approach

The first frequency-to-voltage converter used a reciprocal frequency counter.

Instead of counting the number of input cycles during a fixed gate interval, the FPGA measured the number of 125 MHz FPGA clock cycles occurring during 16 complete periods of the input signal.

If N clock ticks are measured over $M = 16$ input periods, then

$$f_{\text{in}} = \frac{M f_{\text{clk}}}{N}, \quad (8)$$

where

$$f_{\text{clk}} = 125 \text{ MHz}. \quad (9)$$

Thus,

$$f_{\text{in}} = \frac{16 \times 125 \times 10^6}{N}. \quad (10)$$

The calculated frequency was then converted into a DAC code.

4 Limitation of the Reciprocal Counter

The reciprocal counter works well when the input frequency is approximately constant during the measurement interval.

However, the counter must wait for 16 complete input periods before producing a new result.

The measurement time is

$$T_{\text{meas}} = \frac{16}{f_{\text{in}}}. \quad (11)$$

At 20 MHz,

$$T_{\text{meas}} = \frac{16}{20 \times 10^6} = 0.8 \mu\text{s}. \quad (12)$$

During this interval, a rapidly frequency-modulated signal may change its instantaneous frequency.

Therefore, the measured quantity is effectively related to the frequency averaged over the measurement interval rather than a truly instantaneous value.

The problem can be summarized as

$$\boxed{\text{Rapid FM} \rightarrow \text{finite-period measurement} \rightarrow \text{averaging and latency.}} \quad (13)$$

Reducing the number of measured periods would increase the update rate, but it would also reduce frequency resolution and increase sensitivity to timing quantization and noise.

A different frequency-estimation method was therefore implemented.

5 Quadrature Frequency Discriminator

The new converter estimates frequency from the evolution of the *phase* of the signal rather than from the time required to observe multiple complete periods.

The architecture is

$$\boxed{\text{ADC} \rightarrow I/Q \text{ Mixer} \rightarrow \text{LPF} \rightarrow \text{Phase Discriminator} \rightarrow \Delta f \rightarrow \text{Voltage} \rightarrow \text{DAC}.} \quad (14)$$

This approach provides much faster tracking of changing frequency.

6 Digital I/Q Downconversion

Consider an input signal

$$x(t) = A \cos [2\pi f_c t + \phi(t)]. \quad (15)$$

Its instantaneous frequency is

$$f_{\text{in}}(t) = f_c + \frac{1}{2\pi} \frac{d\phi(t)}{dt}. \quad (16)$$

Therefore,

$$\Delta f(t) = \frac{1}{2\pi} \frac{d\phi(t)}{dt}. \quad (17)$$

The FPGA generates two 20 MHz digital reference signals in quadrature:

$$r_I(t) = \cos(2\pi f_c t), \quad (18)$$

$$r_Q(t) = \sin(2\pi f_c t). \quad (19)$$

The ADC signal is multiplied by both references:

$$I_{\text{raw}}(t) = x(t)r_I(t), \quad (20)$$

$$Q_{\text{raw}}(t) = x(t)r_Q(t). \quad (21)$$

Using the product-to-sum identity,

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]. \quad (22)$$

Thus, mixing a signal near 20 MHz with the 20 MHz reference produces a low-frequency difference component and a high-frequency sum component near 40 MHz.

For example,

$$20.1 \text{ MHz} - 20 \text{ MHz} = 100 \text{ kHz}, \quad (23)$$

while

$$20.1 \text{ MHz} + 20 \text{ MHz} = 40.1 \text{ MHz}. \quad (24)$$

The low-pass filter removes the high-frequency mixing product.

7 I/Q Low-Pass Filter

A digital low-pass filter is applied independently to the I and Q channels.

The implemented filter has an approximate -3 dB bandwidth of

$$\boxed{f_{\text{LPF}} \approx 1.138 \text{ MHz.}} \quad (25)$$

The purpose of this filter is not to pass the original 20 MHz carrier.

The carrier has already been translated to baseband by the digital mixer. Therefore, the LPF needs only to preserve the desired baseband frequency/phase variations while suppressing the high-frequency mixer product.

For the present experiment, modulation frequencies up to approximately 100 kHz are expected to remain nearly flat. Higher modulation frequencies can also be recovered, but attenuation and phase shift increase as the modulation frequency approaches the effective bandwidth of the processing chain.

8 Complex Baseband Signal

After I/Q mixing and filtering, the signal can be represented as

$$z[n] = I[n] + jQ[n]. \quad (26)$$

The phase of this complex signal contains the frequency information.

A direct method would calculate

$$\phi[n] = \text{atan2}(Q[n], I[n]) \quad (27)$$

and then calculate

$$\Delta\phi[n] = \phi[n] - \phi[n-1]. \quad (28)$$

However, an efficient implementation can obtain the phase difference directly using a complex-conjugate product.

9 Complex-Conjugate Phase Discriminator

The current complex sample is multiplied by the conjugate of the previous sample:

$$p[n] = z[n]z^*[n-1]. \quad (29)$$

Since

$$z[n] = I[n] + jQ[n], \quad (30)$$

and

$$z^*[n-1] = I[n-1] - jQ[n-1], \quad (31)$$

their product is

$$p[n] = R[n] + jJ[n], \quad (32)$$

where

$$R[n] = I[n]I[n-1] + Q[n]Q[n-1], \quad (33)$$

and

$$J[n] = Q[n]I[n-1] - I[n]Q[n-1]. \quad (34)$$

The phase change is then

$$\boxed{\Delta\phi[n] = \text{atan2}(J[n], R[n])}. \quad (35)$$

This gives the phase rotation between consecutive complex samples.

10 CORDIC Phase Calculation

The atan2 operation is implemented using an FPGA-suitable fixed-point phase calculation, such as a CORDIC (Coordinate Rotation Digital Computer) architecture.

CORDIC is useful in FPGA systems because trigonometric functions can be calculated primarily using shifts, additions, and simple control logic, rather than general floating-point arithmetic.

The phase discriminator therefore converts

$$(R, J) \quad (36)$$

into

$$\Delta\phi. \quad (37)$$

11 Phase-to-Frequency Conversion

Frequency is the time derivative of phase:

$$f = \frac{1}{2\pi} \frac{d\phi}{dt}. \quad (38)$$

In discrete time,

$$\Delta f[n] = \frac{F_s}{2\pi} \Delta\phi[n], \quad (39)$$

where F_s is the effective sampling/update frequency at the phase discriminator.

The implemented design produces a frequency estimate at approximately

$$\boxed{F_{\text{update}} = 5 \text{ MHz.}} \quad (40)$$

Therefore a new frequency estimate is available approximately every

$$T_{\text{update}} = \frac{1}{5 \times 10^6} = 200 \text{ ns.} \quad (41)$$

This is significantly different from the reciprocal counter because the discriminator does not need to wait for 16 complete periods before obtaining a new estimate.

12 Frequency-to-Voltage Scaling

The frequency deviation is converted to voltage using

$$V_{\text{out}} = K_{FV} \Delta f, \quad (42)$$

with

$$K_{FV} = 10 \text{ pV/Hz}. \quad (43)$$

Therefore,

$$\Delta f = 100 \text{ kHz} \quad (44)$$

corresponds to

$$V_{\text{out}} = 100000 \times 10^{-5} = 1 \text{ V}. \quad (45)$$

Similarly,

$$\Delta f = -100 \text{ kHz} \quad (46)$$

corresponds to approximately

$$V_{\text{out}} = -1 \text{ V}. \quad (47)$$

The desired transfer characteristic is therefore

Table 2: Designed frequency-to-voltage conversion.

Input frequency	Frequency deviation	Output voltage
19.900 MHz	-100 kHz	approximately -1 V
19.950 MHz	-50 kHz	approximately -0.5 V
20.000 MHz	0	approximately 0 V
20.050 MHz	+50 kHz	approximately +0.5 V
20.100 MHz	+100 kHz	approximately +1 V

The digital result is limited to the signed 14-bit DAC range to avoid digital overflow.

13 Recovery of a Frequency-Modulated Signal

Consider a sinusoidally frequency-modulated input:

$$f_{\text{in}}(t) = 20 \text{ MHz} + 100 \text{ kHz} \sin(2\pi f_m t). \quad (48)$$

Therefore,

$$\Delta f(t) = 100 \text{ kHz} \sin(2\pi f_m t). \quad (49)$$

Applying the F-to-V sensitivity gives

$$\begin{aligned}
V_{\text{out}}(t) &= (10 \text{ }\mu\text{V/Hz}) (100 \text{ kHz}) \sin(2\pi f_m t) & (50) \\
&= 1 \text{ V} \sin(2\pi f_m t). & (51)
\end{aligned}$$

Hence,

$$V_{\text{out}}(t) \approx 1 \text{ V} \sin(2\pi f_m t). \quad (52)$$

Thus, the modulation waveform can be observed directly on an oscilloscope connected to Red Pitaya OUT2.

14 FPGA Signal Flow

The complete digital signal chain is shown below.

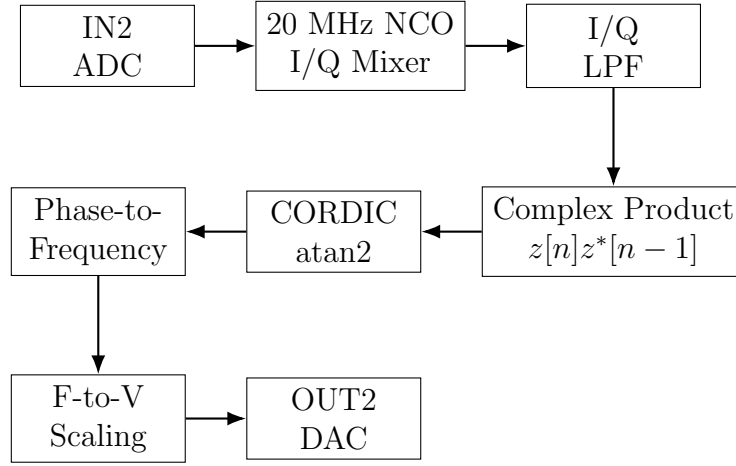


Figure 1: Digital frequency-to-voltage converter implemented inside the FPGA.

The important point is that the external signal entering IN2 is processed entirely digitally after ADC conversion.

No external analog mixer is required for this implementation.

15 FPGA Development and Build

The new implementation was developed inside the existing frequency-to-voltage project environment.

The original reciprocal-counter implementation was preserved and was not overwritten.

Separate HDL/build files were created for the I/Q frequency discriminator.

The final bitstream was generated as

```
build/iq_frequency_to_voltage.bit
```

using Vivado 2020.1.

The FPGA implementation completed successfully.

The reported timing results were

$$\text{Worst setup slack} = +0.306 \text{ ns}, \quad (53)$$

$$\text{Worst hold slack} = +0.035 \text{ ns}. \quad (54)$$

Both values are positive, indicating that the implemented design passed the reported setup and hold timing requirements.

16 FPGA Resource Utilization

The reported resource usage was approximately

Table 3: Reported FPGA resource usage.

Resource	Used
LUTs	4296
Registers	4331
DSP blocks	7
BRAM	0

The DSP blocks are particularly useful for the digital multiplications required by I/Q mixing and complex signal processing.

17 Expected Bandwidth and Performance

The implemented I/Q low-pass filter has a reported -3 dB bandwidth of approximately

$$1.138 \text{ MHz}. \quad (55)$$

The frequency estimate is updated at approximately

$$5 \text{ MHz}, \quad (56)$$

or every

$$200 \text{ ns}. \quad (57)$$

Simulation indicated an approximately flat response from DC to 100 kHz modulation frequency.

The system remained usable at higher modulation frequencies with increasing roll-off. A simulation at a modulation frequency of 500 kHz with a frequency deviation of ± 100 kHz produced approximately 0.906 V peak rather than the ideal 1 V peak.

This reduction is consistent with bandwidth-dependent attenuation in the digital processing chain.

18 Experimental Connection

For testing the new digital F-to-V converter, the frequency-modulated RF signal should be connected directly to Red Pitaya IN2.

The required connection is

$$\boxed{\text{FM Signal} \rightarrow \text{Red Pitaya IN2} \rightarrow \text{FPGA F-to-V} \rightarrow \text{Red Pitaya OUT2} \rightarrow \text{Oscilloscope.}} \quad (58)$$

For example, a useful initial test condition is

$$f_c = 20 \text{ MHz}, \quad (59)$$

$$f_m = 1 \text{ kHz}, \quad (60)$$

$$\Delta f = \pm 50 \text{ kHz}. \quad (61)$$

The expected output is approximately a 1 kHz sinusoid with

$$V_{\text{peak}} \approx 0.5 \text{ V}, \quad (62)$$

or approximately

$$V_{\text{pp}} \approx 1 \text{ V}. \quad (63)$$

The modulation frequency can then be increased gradually while observing the recovered waveform.

19 Important Difference Between the Two Designs

The two F-to-V implementations operate on fundamentally different principles.

Table 4: Comparison of the two frequency measurement methods.

	Reciprocal Counter	I/Q Frequency Discriminator
Measured quantity	Time over multiple periods	Phase change of complex signal
Measurement principle	$f = M f_{\text{clk}} / N$	$\Delta f = (F_s / 2\pi) \Delta \phi$
Update mechanism	Wait for 16 periods	Continuous phase evolution
Rapid FM	Can average changes over measurement window	Designed to follow changing instantaneous frequency
Carrier reference	Not required for basic absolute-frequency measurement	Uses a digital 20 MHz reference
Output	Absolute-frequency-derived voltage	Frequency-deviation-derived voltage

For rapidly changing FM signals around a known carrier, the quadrature discriminator is therefore better suited to recovering the modulation waveform.

20 Limitations

The present implementation has several important limitations.

First, the discriminator is designed for a carrier close to 20 MHz. A sufficiently large carrier offset may move the signal outside the intended baseband processing range.

Second, the output is designed for approximately ± 100 kHz frequency deviation. Larger deviations cause the output to approach or reach the DAC limits.

Third, the digital LPF and subsequent processing impose a finite modulation bandwidth. The present design is approximately flat through the intended DC–100 kHz range, while higher-frequency modulation experiences increasing attenuation.

Fourth, the current implementation does not contain explicit missing-carrier detection. If the RF input disappears or becomes too small, the phase estimate may no longer represent a meaningful frequency.

Finally, the absolute voltage calibration and practical high-frequency performance must be verified experimentally using the Red Pitaya DAC and laboratory instrumentation.

21 Conclusion

A digital frequency-to-voltage converter based on an I/Q frequency discriminator was successfully implemented for the Red Pitaya STEMLab 125-14 FPGA.

The previous reciprocal-counter implementation measured frequency by counting FPGA clock ticks across 16 complete input periods. Although appropriate for relatively stationary signals, this introduces a finite measurement window that is undesirable when attempting to reproduce rapidly varying frequency modulation.

The new implementation digitally downconverts the signal using a 20 MHz quadrature reference, filters the resulting I/Q signals, calculates phase increment using a complex-conjugate product, and converts the phase increment into instantaneous frequency deviation.

The resulting relationship is approximately

$$V_{\text{out}} = (10 \text{ } \mu\text{V/Hz}) (f_{\text{in}} - 20 \text{ MHz}). \quad (64)$$

Thus, a sinusoidal frequency deviation around the 20 MHz carrier is recovered as a corresponding sinusoidal voltage waveform at Red Pitaya OUT2.

The final FPGA bitstream is

```
build/iq_frequency_to_voltage.bit
```

and the design successfully passed FPGA timing implementation.